

Impact of Crystal Geometry Simplification on HPGe Detector Response: A Monte Carlo Study

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Abstract

Accurate gamma-ray spectrometry relies on precise detector calibration, which is essential for applications in radiological protection and nuclear engineering. Although experimental calibration remains the reference approach, its implementation is often limited by the high cost and limited availability of certified radioactive sources. Monte Carlo simulations provide a flexible and reliable alternative for simulating detector response and determining efficiency curves [Conti *et al.*, 2013]. Commercial HPGe detectors typically feature rounded crystal edges resulting from the crystal manufacturing process. However, these geometric features are frequently neglected in computational models because their representation considerably increases modeling complexity. The impact of this geometric simplification on the simulated full-energy peak efficiency has received limited systematic investigation. This work presents two computational models of a GC2520 HPGe detector developed using the MCNP6 code [Goorley *et al.*, 2016]. The first employs a simplified crystal geometry with flat peripheral surfaces [Konstantinova *et al.*, 2021], whereas the second explicitly incorporates the rounded crystal edges produced during crystal fabrication. Experimental absolute full-energy peak efficiencies are measured with a point source of ¹⁵²Eu, and full-energy peak efficiency curves are generated over a broader gamma-ray energy range for both detector models to assess the influence of crystal edge geometry. With all other detector parameters kept unchanged, differences between the resulting

efficiency curves isolate the effect of crystal edge geometry on the simulated detector response. This difference remained close to 5% across the entire energy range investigated, in clear contrast with the substantially larger and strongly energy-dependent discrepancy against the experimental data observed at low energies. Although both models deliberately excluded the detector's dead layer, the direction of this low-energy discrepancy does not fully support the dead layer as its sole origin, indicating that further investigation is required. At high energies, both models achieved comparable agreement with the experimental data, indicating that the simplified geometry offers a computationally more efficient alternative without a significant loss of accuracy for such applications.

Keywords: HPGe Detector; Monte Carlo Simulation; Detector Modeling.

1.- INTRODUCTION

Accurate quantification of radionuclide activity by gamma-ray spectrometry with high-purity germanium (HPGe) detectors is a cornerstone of radiological protection, environmental monitoring and nuclear engineering applications, since the reliability of the results depends directly on accurate knowledge of the full-energy peak efficiency over the energy range of interest.

The full-energy peak efficiency of a detector is traditionally obtained through experimental calibration with a set of certified point sources spanning the relevant energy range. Although this remains the reference approach, it is often constrained by the high cost, limited availability and restricted energy coverage of certified radioactive sources, particularly when several measurement geometries must be calibrated.

Monte Carlo simulation, and in particular the MCNP6 code, provides a flexible and reliable alternative to purely experimental calibration [Goorley *et al.*, 2016], allowing the full-energy peak efficiency to be estimated for arbitrary source-detector geometries and energies once an accurate computational model of the detector has been established. Commercial HPGe detectors, however, typically present rounded crystal edges that result from the crystal manufacturing process, and this geometric feature can influence the effective active volume

seen by incident photons, particularly at low energies and near the crystal periphery. This detail is commonly simplified in computational models, with the rounded crystal edges being replaced by flat peripheral surfaces [Konstantinova *et al.*, 2021]. Although the rounded-edge radius has also been optimized directly against experimental scan data [Haj-Heidari *et al.*, 2016], its explicit representation considerably increases the complexity of the geometry. Consequently, the impact of this simplification on the simulated full-energy peak efficiency has received limited systematic investigation.

More broadly, simulated efficiencies are also sensitive to other detector-geometry parameters, such as the dead layer thickness and the crystal-to-endcap distance, which may differ from manufacturer specifications and often require iterative optimization against experimental data [Agarwal *et al.*, 2011; Dokania *et al.*, 2014; Gealone *et al.*, 2026]. Karamanis *et al.* [2002], for instance, showed that reproducing the measured efficiencies required adjustments to detector-geometry parameters, including the crystal-to-window distance and dead layer thickness. A comparable adjustment was reported for a well-type HPGe detector, for which the combined dead plus transition layer thickness was determined iteratively for each crystal surface [Espirito Santo *et al.*, 2011].

Establishing the conditions under which the simplified geometry remains an acceptable approximation is therefore relevant both for routine detector calibration and for benchmarking studies. The objective of this work was to evaluate the influence of crystal edge geometry simplification on the simulated full-energy peak efficiency of a GC2520 HPGe detector over the extended gamma-ray energy range provided by the multiple emissions of ^{152}Eu , using two MCNP6 computational models that reproduce the manufacturer's crystal dimensions and differ only in the representation of the crystal periphery, one with simplified flat surfaces [Konstantinova *et al.*, 2021] and one with the rounded edges produced during fabrication.

2.- MATERIALS AND METHODS

This section describes the HPGe detector and the point-source measurements used as reference, the Monte Carlo models developed to represent the detector, and the procedure used to generate and compare full-energy peak efficiency curves against ^{152}Eu measurements over its extended gamma-ray energy range.

2.1.- HPGe Detector and Experimental Measurements

Experimental measurements were performed with a coaxial p-type Canberra GC2520 HPGe detector (crystal diameter 62.00 mm; crystal length 34.60 mm; relative efficiency $\geq 25\%$; energy resolution ≤ 2.0 keV FWHM at 1332 keV), with a 3.00 in (76.2 mm) diameter x 5.25 in (133.4 mm) length end cap, coupled to standard gamma-ray spectroscopy electronics. Absolute full-energy peak efficiencies were obtained at a fixed source-to-detector distance, positioned on the detector symmetry axis, using a calibrated ^{152}Eu point source, whose multiple gamma-ray emissions provide efficiency data over an extended energy range. For each gamma-ray energy, the absolute full-energy peak efficiency, ϵ_{exp} , was calculated directly by the detector's acquisition and analysis software Genie 2000, from the net peak area, the source activity decay-corrected to the measurement date, the gamma-ray emission probability, and the counting live time. The ^{152}Eu point source had a calibrated activity of 60.277 kBq, and the measurement was carried out with a 24 h counting live time, with the source positioned at 20.456 cm from the detector using support stands to reduce the dead time, which reached 0.84%. This comparatively large source-to-detector distance also minimizes true coincidence summing effects associated with the cascade emissions of ^{152}Eu , which become significant mainly in close counting geometries [Karamanis *et al.*, 2002]. The combined standard uncertainty of ϵ_{exp} , as reported by the acquisition and analysis software, accounts for the uncertainties associated with the net peak area, the source activity, and gamma-ray emission probability.

2.2.- Monte Carlo Detector Models (MCNP6)

Two computational models of the GC2520 detector were developed with the MCNP6 Monte Carlo code [Goorley *et al.*, 2016], both reproducing the manufacturer's crystal dimensions

(62.00 mm diameter, 34.60 mm length) and the central axial bore corresponding to the inner electrode contact, and sharing an identical source term. The two models differ only in the representation of the crystal periphery and of the electrode bore: in the first (basic) model, the lateral and front surfaces of the crystal, as well as the electrode bore, meet at sharp, square edges; in the second (advanced) model, additional cells were introduced at the crystal periphery and at the electrode bore to reproduce the rounded edges produced during fabrication, as shown in Figure 1 (left) and (right).

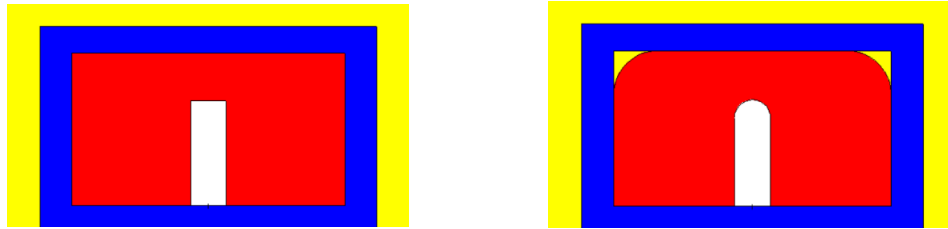


Figure 1.- Cross-sectional views of the (left) basic and (right) advanced MCNP6 detector models.

For both models, the electrode dimensions were determined in two steps: the initial estimate proposed by Conti *et al.* [2013] was first adopted and then refined iteratively, by comparing the simulated full-energy peak efficiency at the highest-energy ^{152}Eu emission line (1.40795 MeV) with the corresponding experimental value and selecting the size that minimized the relative discrepancy. In both models, photon and electron transport was considered (MCNP6 coupled mode P E), and the full-energy peak efficiency was estimated using the pulse-height tally (F8). Simulations were run with 1×10^8 source histories (NPS), which was sufficient to keep the relative statistical uncertainty of the tally below 1% at the energies of interest. All simulations were performed on an Intel Core i3-10100T processor with 8 GB of RAM, using 4 cores/8 threads, and the resulting computational times are reported in Table 3. Neither model included the detector's dead layer, since its precise determination would require an additional iterative calibration procedure; this choice also allowed the comparison to be performed on the nominal geometry rather than on the real active volume. Introducing the dead layer removes an inactive outer shell of germanium from the sensitive volume, so the simulated efficiency is expected to decrease once it is included, particularly at low gamma-ray energies, where photon interactions occur closer to the crystal surface.

2.3.- Efficiency Curve Generation and Comparison

Full-energy peak efficiency curves were generated for both detector models over the extended energy range provided by the multiple emission lines of ^{152}Eu , using the crystal dimensions described in Section 2.2 and keeping all other detector parameters unchanged, and were compared with the experimental efficiencies obtained with the ^{152}Eu point source (Section 2.1). Since the two models differ only in the geometric representation of the crystal periphery and the electrode bore, the relative difference between the resulting curves isolates the combined effect of these simplifications across the range covered by ^{152}Eu , independently of other potential sources of discrepancy between simulation and experiment.

3.- RESULTS

At the highest-energy ^{152}Eu emission line (1.40795 MeV), used as the reference for the electrode-size iteration, the initial electrode estimate proposed by Conti *et al.* [2013] yielded a discrepancy between the simulated (advanced model) and experimental efficiencies of 4.49%. Refining the electrode size iteratively against the advanced model reduced this discrepancy to 3.66% at the same energy, and the refined electrode size was subsequently adopted in both detector models.

Full-energy peak efficiencies were determined at the eleven ^{152}Eu emission energies, from 121.78 keV to 1407.95 keV. Tables 1 and 2 present, respectively for the basic and advanced models, the experimental and simulated efficiencies obtained with the refined electrode size, together with the relative discrepancy at each energy. The relative statistical uncertainty of the simulated efficiencies remained below 1% at all energies, in agreement with the criterion adopted in Section 2.2.

Table 1.- Simulated (basic model) and experimental full-energy peak efficiencies for ^{152}Eu .

Energy [MeV]	Experimental Efficiency ($\times 10^{-3}$)	Simulated – Basic Efficiency ($\times 10^{-3}$)	Discrepancy (%)
0.12178	3.660 $\pm$ 0.002	2.940 $\pm$ 0.017	19.54

0.24469	2.540±0.003	2.270±0.015	10.52
0.34427	1.780±0.001	1.650±0.013	7.08
0.41111	1.500±0.004	1.400±0.004	6.54
0.44398	1.370±0.004	1.300±0.004	4.91
0.77889	0.800±0.001	0.783±0.003	2.12
0.86732	0.728±0.002	0.713±0.003	2.11
0.96401	0.658±0.001	0.650± 0.003	1.17
1.08578	0.593±0.001	0.589±0.002	0.68
1.11202	0.581±0.001	0.576±0.002	0.78
1.40795	0.464±0.001	0.470±0.002	1.25

Table 2.- Simulated (advanced model) and experimental full-energy peak efficiencies for ^{152}Eu .

Energy [MeV]	Experimental Efficiency (x10 ⁻³)	Simulated – Advanced Efficiency (x10 ⁻³)	Discrepancy (%)
0.12178	3.660±0.002	2.770±0.017	24.45
0.24469	2.540±0.003	2.160±0.015	15.03
0.34427	1.780±0.001	1.570±0.013	11.98
0.41111	1.500±0.004	1.340±0.004	10.94
0.44398	1.370±0.004	1.240±0.003	9.44
0.77889	0.800±0.001	0.745±0.003	6.83
0.86732	0.728±0.002	0.681±0.003	6.52
0.96401	0.658±0.001	0.620±0.002	5.84
1.08578	0.593±0.001	0.560±0.002	5.51
1.11202	0.581±0.001	0.550±0.002	5.29
1.40795	0.464±0.001	0.447±0.002	3.66

Table 3 presents the relative difference between the simulated efficiencies of the two models, computed using the advanced model as reference, which isolates the effect of the crystal edge geometry independently of the experimental comparison, together with the computational time of each simulation.

Table 3.- Relative difference between the basic and advanced simulated efficiencies for ^{152}Eu , and their respective computational times.

Energy [MeV]	Basic-Advanced Difference (%)	Computational Time – Basic [min]	Computational Time – Advanced [min]
0.12178	6.50	164.60	192.57
0.24469	5.31	81.93	96.72
0.34427	5.57	60.28	78.59
0.41111	4.95	562.43	714.11
0.44398	5.00	548.34	705.34
0.77889	5.05	561.84	801.10
0.86732	4.72	573.68	838.86
0.96401	4.96	584.15	883.43
1.08578	5.11	604.27	932.33
1.11202	4.77	605.09	955.03
1.40795	5.09	636.22	1055.10

Figure 2 shows the experimental and simulated efficiency curves over the full energy range investigated. Both simulated curves reproduce the overall shape of the experimental curve, with a steep decrease at low energies followed by a more gradual decline. The largest discrepancies occur at the lowest energies, reaching 24.45% (advanced, Table 2) and 19.54% (basic, Table 1) at 121.78 keV, and decrease progressively with energy.

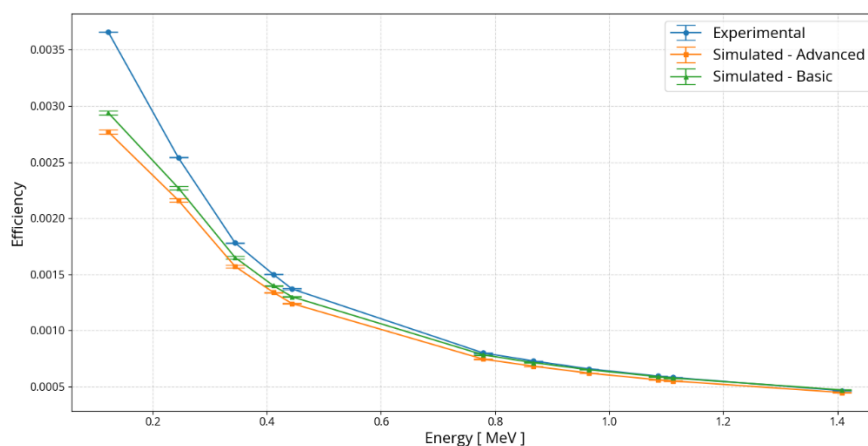


Figure 2.- Experimental and simulated (basic and advanced models) full-energy peak efficiency curves for ^{152}Eu .

Tables 1 and 2 also detail the high-energy region of the spectrum (above ~ 750 keV), where the discrepancy against the experimental data drops substantially with respect to the low-energy region, remaining below 7% for the advanced model and below 3% for the basic model and reaching its lowest values, 3.66% and 1.25% respectively, at 1407.95 keV, the energy used as the reference for the electrode-size iteration.

4.- DISCUSSION

The pronounced discrepancies observed at low energies might, in principle, relate to the deliberate exclusion of the detector's dead layer from both models (Section 2.2), since low-energy photons interact preferentially near the crystal surface. However, both simulated efficiencies are lower than the corresponding experimental values at these energies (Tables 1 and 2), not the direction expected from neglecting an inactive surface layer alone, since omitting the dead layer should, if anything, increase rather than decrease the simulated active volume seen by incident photons. This indicates that the low-energy discrepancy is more complex than a single geometric simplification and cannot be conclusively attributed to the dead layer with the present models; clarifying its origin will require new MCNP6 models that explicitly incorporate the dead layer, alongside experimentally validated electrode dimensions, compared directly against the experimental efficiencies reported here. This discrepancy is markedly reduced at higher energies, where photon interactions occur, on average, deeper within the active volume, consistent with the substantially better agreement observed there (Tables 1 and 2).

Averaged over the eleven ^{152}Eu energies, the discrepancy against experiment is 5.2% for the basic model (Table 1) and 9.6% for the advanced model (Table 2), the basic model being closer to experiment at every energy by an amount consistent with the nearly constant $\sim 5\%$ difference between the two simulated curves (Table 3). This does not indicate that the simplified geometry is a more accurate representation of the detector: the electrode size was

calibrated against the advanced model using the highest-energy line (1.40795 MeV) and then kept unchanged in the basic model (Section 2.2), so its incidental reduction of the basic model's discrepancy is a by-product of the calibration procedure, not evidence of superior accuracy. A more meaningful indicator of this geometric simplification's influence is the relative difference between the two simulated curves themselves (Table 3). Since the models differ only in the crystal periphery and the electrode bore, this difference isolates the intrinsic effect of neglecting the rounded edges at both locations from other sources of discrepancy such as the dead layer, and it stays essentially constant across the energy range studied, indicating a systematic bias in the simulated absolute efficiency rather than one concentrated in a specific energy region. Because both features were simplified simultaneously, their individual contributions to the $\sim 5\%$ bias cannot be separated here; doing so would require models that vary one feature at a time.

Despite this constant offset, the discrepancy of each model against the experimental data decreases markedly at high energies (Tables 1 and 2), so in this region the choice between the basic and advanced geometries has little practical effect on accuracy, favoring the simplified model, an advantage reinforced by its lower computational cost (Table 3).

5.- CONCLUSIONS

This work fulfilled its objective of evaluating the influence of crystal periphery and electrode-bore geometry simplification on the simulated full-energy peak efficiency of a GC2520 HPGe detector, through two MCNP6 models that differ only in the representation of these two features.

The main finding is that simplifying the crystal periphery and electrode-bore geometry introduces a systematic bias of approximately 5% in the simulated absolute efficiency, essentially constant across the energy range studied (Table 3), and markedly smaller than the strongly energy-dependent discrepancy against the experimental data observed at low energies (Section 4). This low-energy discrepancy was initially expected to result mainly from the detector's dead layer, deliberately excluded from both models, but the direction of

the observed differences does not fully support this explanation, so its origin should be regarded as an open question. Confirming whether geometric simplification and the dead layer indeed act as distinct, separable sources of discrepancy in HPGe detector simulations will require future models that explicitly incorporate the dead layer.

In addition, at high energies the discrepancy against the experimental data is small for both models (below 3% for the basic model and below 7% for the advanced model, Tables 1 and 2); the basic geometry achieves comparable accuracy while requiring substantially less computational time, about 40% less at the highest-energy line (Table 3), making it a practical choice for high-energy applications. These results highlight the importance of matching the level of geometric detail to the accuracy required by the application.

Future work should address some limitations of this study. Neither model included the detector's dead layer, and other detector components and material layers were not fully characterized. Incorporating these features in both models, together with experimentally validated internal-electrode dimensions and appropriate geometry-optimization procedures, is necessary to assess whether the observed $\sim 5\%$ offset persists after these additional detector parameters are properly characterized and modeled, and to clarify the low-energy discrepancies discussed above. Because the crystal periphery and electrode bore were simplified simultaneously, their individual contributions could not be separated in the present study; models that vary one feature at a time would allow this distinction. Finally, experimental calibration remains indispensable as the reference benchmark for any Monte Carlo model, and extending this comparison to additional detectors and geometries would help establish how broadly these conclusions apply.

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REFERENCES

- Agarwal C; Chaudhury S; Goswami A; Gathibandhe M. (2011). Full energy peak efficiency calibration of HPGe detector for point and extended sources using Monte Carlo code. *Journal of Radioanalytical and Nuclear Chemistry* 287: 701-708.
- Conti CC; Salinas ICP; Zylberberg H. (2013). A detailed procedure to simulate an HPGe detector with MCNP5. *Progress in Nuclear Energy* 66: 35-40.
- Dokania N; Singh V; Mathimalar S; Nanal V; Pal S; Pillay RG. (2014). Characterization and modeling of a low background HPGe detector. *arXiv:1311.4779*.
- Espirito Santo AS; Conti CC; Wasserman FG. (2011). Calibration and simulation of a HPGe well detector using Monte Carlo computer code. In: *International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering (M&C 2011)*, Rio de Janeiro, Brazil.
- Gealone DAT; Balderas CV; Hila FC; Bauyon-Isidro MM; Racho JMD; Jagonoy AM; Astronomo AA. (2026). Optimized Monte Carlo modeling of HPGe detector efficiency with true coincidence summing corrections using PHITS and MCNP5. *Applied Radiation and Isotopes* 238: 112900.
- Goorley T *et al.* (2016). Features of MCNP6. *Annals of Nuclear Energy* 87: 772-783.
- Haj-Heidari MT; Safari MJ; Afarideh H; Rouhi H. (2016). Method for developing HPGe detector model in Monte Carlo simulation codes. *Radiation Measurements* 88: 1-6.
- Karamanis D; Lacoste V; Andriamonje S; Barreau G; Petit M. (2002). Experimental and simulated efficiency of a HPGe detector with point-like and extended sources. *Nuclear Instruments and Methods in Physics Research A* 487: 477-487.
- Konstantinova M; Germanas D; Gudelis A; Plukis A. (2021). Efficiency calibration of high-purity germanium detector using Monte Carlo simulations including coincidence-summing corrections: volume source case. *Lithuanian Journal of Physics* 61: 66-73.